

Direct gradient-based topology optimization of differentiable scaffold geometries under physicochemical ageing

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Abstract

Load-bearing scaffold structures are exposed to a variety of application-dependent conditions under which their local material properties change over time. However, this change of properties is largely determined by the lattice geometry itself. In the present paper, a direct gradient-based topology optimization framework is demonstrated in which a differentiable implicit lattice geometry is coupled to geometry-dependent evolving material states. The geometry is represented by integrable phase potentials, a wall-thickness field, and a macroscopic topology mask. The environmental exposure, state evolution, local material properties, and resolved mechanics are combined within one differentiable chain, through which the geometry is optimized with respect to the material state reached under environmental influences. This numerical structure is applied to a thermoelastic loading case, cyclic deterioration, oxidation, photolytic exposure, and the influence stiffness-gaining tissue-deposition, resembling typical load cases in tissue engineering applications. Based on two-dimensional studies, mechanism-specific geometric adaptations and substantial state-mediated changes of the design sensitivity are obtained. In a penalized-interface benchmark, deterioration-aware continuations from three blind reference geometries reduce the aged compliance of volume-matched binary lattices by 7.72% - 8.47% in a tissue-engineering related multiphysical ageing scenario. The paper thus provides a numerical proof of concept for the efficient, direct geometry optimization differentiated through evolving material states under tissue engineering conditions.

1 Introduction

Topology optimization is generally applied to improve the distribution of material according to prescribed loads and constraints. For load-bearing scaffolds, the resulting performance and, thus, the response to imposed external forces and thus its influence in cell ingrowth and proliferation, largely depends on the architecture at both the structural and the cell scale. For instance, local wall orientations and wall thicknesses determine the load paths, while pores and exposed surfaces govern the transport, thermal penetration, and access of environmental media. The geometry therefore influences the initial stiffness as well as the conditions under which the local material state evolves during service.

Under service conditions extending beyond an idealized purely mechanical load case, thermal loading, cyclic stressing, oxidation, photolytic exposure, hydrolytic resorption, and, for the particular case of bioengineering scaffolds, tissue deposition may alter the local stiffness over time. These processes are commonly evaluated after a geometry has been designed or are introduced through material fields that remain partly separated from the geometric description. Such a sequential treatment remains suitable where the exposure is prescribed independently of the structure. A stronger interaction is obtained where the current geometry controls the exposure itself. For the particular application of scaffolds, changes in wall thickness may alter oxygen transport, changes in the load path may alter cyclic deterioration, and the pore architecture may determine the access of migrating cells to nutrients and, thus, the degree of tissue deposition. The aged or cultured mechanical response must consequently be considered during the selection of the geometry.

Structural optimization under ageing therefore requires a continuous relation between the geometry, exposure, state evolution, and mechanical properties. Within this relation, a coherent lattice description has to be retained while the service-dependent state is evaluated on the current geometry, after which the sensitivity of the end-state performance has to be returned through the complete chain to the geometric design variables. A common formulation is additionally required through which different ageing and growth laws can be integrated without changes to the underlying optimization procedure. The simultaneous differentiation of a coherent lat-

tice geometry and its geometry-dependent evolving material state constitutes the central engineering problem considered within the present paper.

2 State of the art

Topology optimization is widely applied to determine load-bearing material structures, for which the available approaches differ in the representation of the design. Within element-density methods, high geometric freedom is obtained through homogenization-based material distributions [1], while mature design procedures are summarized in standard treatments [2]. The principal topology-optimization families have consequently been compared at the methodological level [3]. Since density methods provide an intermediate material field, geometric alternatives with an explicit boundary description have developed in parallel. In level-set methods, the structural boundary is described implicitly [4] and evolved using sensitivity information [5], whereas identifiable components are retained in feature-mapping methods through moving morphable geometries [6] or differentiable projections of discrete elements [7]. The resulting engineering distinction concerns the design output. For density-based methods, conversion into a manufacturable geometry is required, while information on boundaries, members, and feature dimensions is retained throughout optimization by direct descriptions. Lattice optimization has developed along both routes.

For large structural domains, the computational effort is reduced by two-scale methods through separation of effective-property optimization from the realization of the fine architecture. Optimized composite microstructures can be projected along their lamination directions in a post-treatment [8], while high-resolution manufacturable microstructures are reconstructed from the optimized fields by projection-based dehomogenization [9]. The broader development and its realization problem are reviewed in [10]. This separation makes large domains tractable and requires the preservation of cell continuity, feature size, and load transfer in the reconstructed geometry. A closer connection between the scales has therefore been established through local TPMS grading within a macro-layout [11] or surface-lattice reconstruction after homogenization-based optimization [12]. In other formulations, a larger part of the lattice geometry is included in the optimization. Spatial grading has been realized by explicit topology optimization [13], while stress-constrained designs have been obtained through asymptotic homogenization [14]. Directional information is converted into conforming lattices by field-aligned parameterization [15] and into surface lattices by computational conformal mapping [16]. The remaining separation is further reduced through fully resolved descriptions based on projection-based implicit lattices [17], differentiable Voronoi cells [18], continuous cutting fields [19], or function-based TPMS representations [20]. This direct relation between geometry and objective has also been applied to the optimization of TPMS period and wall thickness for heat dissipation [21] and to the optimization of TPMS-based porous structures through a neural implicit representation [22].

Once the geometry is included in the differentiable design description, a broader range of behavior can be considered without changing the basic optimization principle. An increased flexibility of the design field has been demonstrated through neural topology parameterizations [23] and implicit neural multiphysics representations [24]. Large-displacement structures [25], finite-strain behavior [26], periodic microstructures with tailored homogenized properties [27], and concurrent hollow structures with discrete manufacturable lattices [28] have furthermore been considered through nonlinear formulations. Complementary data-driven inverse-design approaches have been demonstrated for spinodoid metamaterials [29], nonlinear mechanical metamaterials generated through video diffusion [30], and experiment-informed finite-strain spinodal behavior [31]. An increasingly direct control of the wall thickness, cell size, orientation, anisotropy, and pore connectivity is thereby obtained at increased numerical effort when the complete lattice is resolved.

A related line of development concerns the material state reached during service in addition to the initial structure. Internal-variable criteria [32] and fully coupled nonlinear continuum damage [33] have been introduced in damage-aware formulations. Because repeated loading accumulates over the life cycle, fatigue has been incorporated through finite-life constraints [34] and continuous-time high-cycle models [35]. For biodegradable composites, the changing state has been considered through time-dependent degradation [36], uncertain degradation rates [37], and diffusion-driven degradation coupled to structural shape optimization [38]. The same interaction has been considered in scaffold design, where temporary mechanical support has to be balanced against biological constraints [39]. The integration of porous scaffold design with controlled fabrication is reviewed in [40]. Time-dependent bone regeneration has more recently been considered through optimized scaffold porosity [41], while mechanobiological bone ingrowth has been differentiated through a time-dependent level-set formulation [42].

The evolution laws that define these service-dependent states differ according to the underlying mechanism. Stiffness-losing degradation is associated with polymer chain scission and erosion [43]. Whether surface or bulk erosion dominates depends on the relation between diffusion and reaction and on the dimensions of the polymer matrix [44], while bulk-eroding scaffold degradation has been described through hydrolysis and mass transport [45]. Tissue growth changes the material state in the opposite direction because deposition progresses

through curvature-dependent fronts [46] and can be coupled to shear-stimulated growth under perfusion [47]. An enhancement of matrix elaboration in chondrocyte constructs may further be obtained under dynamic loading [48], while the broader evidence for dynamic mechanical compression is assessed in a critical review [49]. The local material properties reached over the life cycle may therefore govern the structural optimum and have to be included during optimization. Although the modelling of environmental ageing and durability of polymer components is well established [50], the coherent integration of multiple physical ageing mechanisms into topology optimization remains challenging. This coupling is particularly relevant for additively manufactured lattices because transport depends on the wall thickness and pore connectivity, cyclic deterioration depends on the evolving load paths, and radiative ageing depends on surface accessibility. A common formulation is consequently required in which the exposure is determined from the geometry, the local properties are changed by the evolving state, and the geometric update is based on the resulting end-state performance. Joint optimization therefore requires the simultaneous differentiability of the coherent lattice geometry and the interlinked geometry-dependent ageing or growth state.

3 Methodology

3.1 Design variables and optimization procedure

The evolving geometries are represented by three coarse fields stored on a control grid that is six times coarser than the analysis mesh in the present study. For the two-dimensional beam considered here, 31×11 control nodes are applied over 180×60 finite elements. The calculations are conducted in two dimensions, while the underlying formulation permits extension to three-dimensional domains. The first field modifies the lattice phase, while the local wall thickness and the macro-topology are defined by the second and third fields, respectively. At every optimization iteration, the fields are interpolated bilinearly to the analysis mesh and transformed into a material distribution through the geometric map of Section 3.2. The element modulus is obtained from the geometric occupancy ρ and the associated penalized stiffness density ρ_s ,

$$\rho_{\text{lat}} = \sigma\left(\frac{t - \delta}{wh}\right), \quad \rho = \tilde{m} \rho_{\text{lat}}, \quad \rho_s = \tilde{m}^{p_{\text{msk}}} \rho_{\text{lat}}^{q_{\text{lat}}}. \quad (1)$$

Here, σ denotes the logistic sigmoid. The exponents are set to $p_{\text{msk}} = 3$ and $q_{\text{lat}} = 1$ in the principal campaigns, which induces a cubic penalty of the macro-topology mask and a linear penalty of the diffuse wall interface. The penalized-interface benchmark uses $q_{\text{lat}} = 3$.

Depending on the applied load case and state coupling, one or more finite element problems are solved on the resulting stiffness distribution. The mechanical objective is the compliance $C = b^T u$, while additional physical-state solves are introduced for the thermal, ageing, and tissue-deposition couplings. The objective and the geometric regularization terms are differentiated with respect to every control value through reverse-mode automatic differentiation. Each wrapped linear equilibrium requires one additional adjoint solve with the same matrix, irrespective of the number of design variables. The raw control fields are updated using Adam [51], after which the prescribed volume is restored by the scalar correction described in Section 3.4. Unless stated otherwise, the studies within the present paper were initialized from the unperturbed uniform lattice, $\delta\boldsymbol{\varphi} = \mathbf{0}$, for which both logit fields are set to zero and are subsequently shifted to the target volume. The material parameters are $E_0 = 1$, $\nu = 0.3$, and $E_{\text{min}} = 10^{-4}$. Since the load resultant, solid modulus, and element edge are normalized to unity, all compliances are nondimensional. The complete sequence is summarized in Fig. 1, including the generic physics-agnostic dependencies and the physics-specific variables applied throughout the paper. Solid arrows denote the forward calculation, dashed arrows the reverse derivative and update, thin arrows internal sub-process flows, and the dash-dotted arrow the dependence of the excitation on the design.

3.2 Geometric representation

The design object is represented by a sheet lattice whose walls follow a periodic mid-surface. The local cell geometry, wall thickness, and spatial extent are described by three fields defined on a control grid with a spacing of six element edges. The walls are given by level lines of the phase field, for which the potential $\Phi_i(\mathbf{x})$ accumulates the phase along direction i . The uniform lattice is defined by $\Phi_i = k_0 x_i$ with $k_0 = 2\pi/\text{period}$, and a nominal period of twelve length units is applied in the present study. Smooth phase corrections are stored on the control grid according to $\Phi_i = k_0 x_i + \delta\varphi_i(\mathbf{x})$, through which the local orientation and spacing are graded. Since the lattice directions are obtained from the gradients of the phase potentials, $\mathbf{J} = \nabla\Phi$, a compatible phase description is retained across neighboring cells and seams or dislocations associated with independently assigned element orientations are suppressed. The periodic lattice function is evaluated as $f = \cos \Phi_1 + \cos \Phi_2$.

The wall thickness results from the symmetric deposition of material around the level set $f = 0$. The unsigned distance is approximated by $\delta = |f|/|\nabla f|$, which constitutes a first-order distance surrogate and

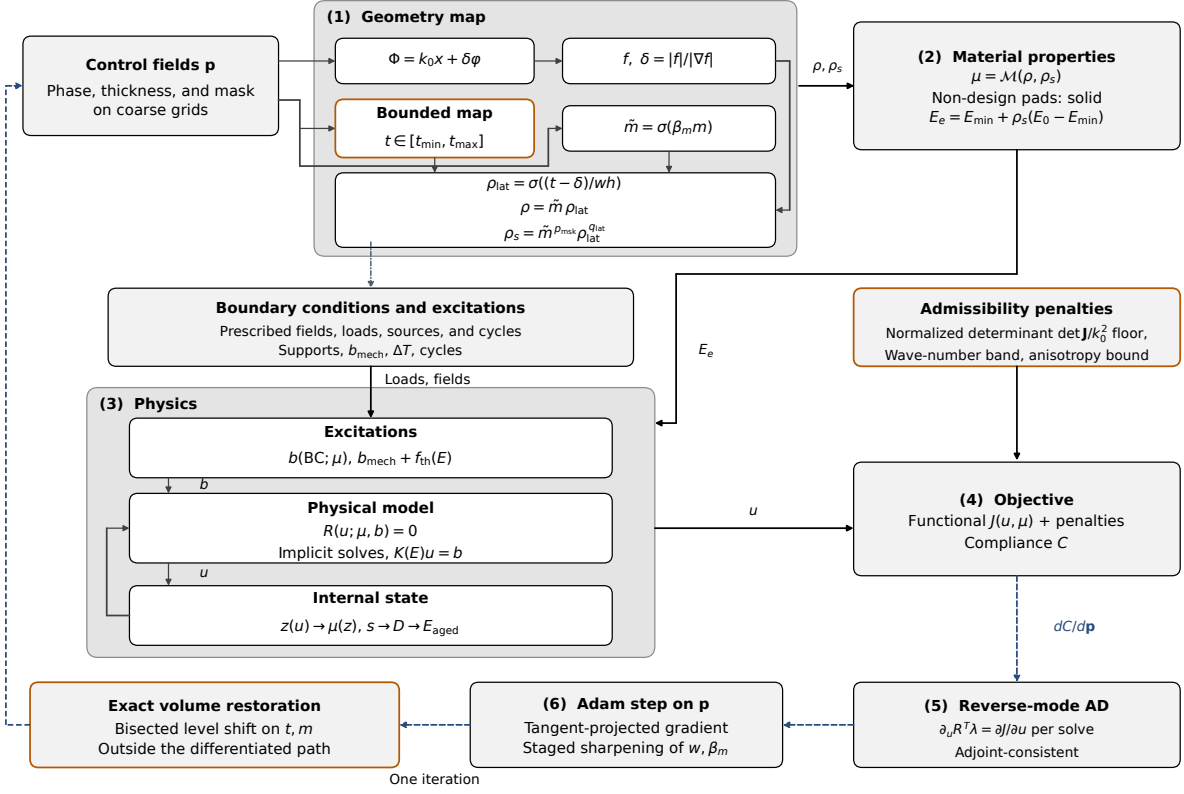


Figure 1: Overview of one optimization iteration, comprising the geometry map, the property map, the physics block with its internal state, the objective, the reverse derivative, the Adam update, and the scalar volume restoration.

coincides with the geometric distance in the vicinity of a regular level set. The gradient magnitude is evaluated analytically through $\mathbf{J}$, smoothed over a quarter period, and floored to retain a finite value at the lattice junctions. Material is assigned through the condition $\delta < t$, thereby defining a wall with the local half thickness t . The thickness field is mapped into $[t_{\min}, t_{\max}]$, such that the half-thickness parameter remains bounded throughout optimization. Complete macroscopic removal of walls is represented by the additional macro-topology control m . The interpolated mask is projected as $\tilde{m} = \sigma(\beta_m m)$ and sharpened from $\beta_m = 4$ to 128 during the continuation. The differentiable occupancy is $\rho = \tilde{m} \sigma((t - \delta)/(wh))$, with the interface half width w reduced from two element sizes to 0.35. The macro-topology is therefore governed by the mask, while the local lattice morphology is governed by the phase and thickness fields.

For regularization of the phase map, three quadratic penalties are applied, comprising a lower target on $\det \mathbf{J}$ against local foldover, an upper target on the row norms of $\mathbf{J}$ against underresolved cells, and a restriction based on the closed-form singular values of the local frame against excessive anisotropy. These soft restrictions retain a smooth optimization problem while constraining the accessible design space. Across the 31 principal final designs and the strip control, the determinant remained positive at every sampled point, thereby confirming local orientation preservation. The nominal targets were exceeded locally, with a minimum normalized determinant of 0.082 compared with the target of 0.25, a maximum frame condition number of 8.7 compared with a target of 4, and a maximum local wave number of 1.31 times its nominal cap. The largest mean penalty residual remained below $6.3 \cdot 10^{-3}$. In all computed scenarios, the load-introduction and support regions are represented by non-design solid pads and included in the prescribed volume. Identical support pads are used for every load case, while the distributed load additionally contains a two-element solid strip along the loaded edge through which the load is applied to a geometrically defined region.

3.3 Geometry-dependent sensitivity

Based on the definitions of Section 3.2, the complete dependency chain can be written as

$$\begin{aligned} \mathbf{p} &\rightarrow (\delta \varphi, t, \tilde{m}) \rightarrow (f, \delta) \\ &\rightarrow (\rho, \rho_s) \rightarrow E_e \rightarrow u \rightarrow C. \end{aligned}$$

The objective therefore constitutes a single differentiable function $C(\mathbf{p})$. Related parameterized and data-driven design descriptions have been introduced for the inverse design of spinodoid metamaterials [29], the

video-diffusion-based inverse design of nonlinear mechanical metamaterials [30], experiment-informed finite-strain inverse design [31], and neural density parameterizations [23]. Within the present formulation, the derivative is evaluated through resolved mechanics on a background-grid projection of the implicit lattice, by which a direct relation between the lattice geometry and the objective is established while the final geometry is retained within the same representation. Smooth numerical regularizations are applied at singular points together with sigmoidal material transitions, bounded thickness and mask mappings, and differentiable penalty functions, through which the local phase-map foldover, underresolution, and excessive anisotropy are penalized. The evaluation of $dC/d\mathbf{p}$ is performed by traversing the chain in reverse through automatic reverse-mode differentiation. Since the equilibrium $u(E)$ is defined implicitly by $K(E)u = b$, a specific derivative rule is required. Differentiation of the discrete equilibrium yields the adjoint relations

$$K\lambda = \frac{\partial C}{\partial u}, \quad \frac{dC}{d\mathbf{p}} = \frac{\partial C}{\partial \mathbf{p}} + \lambda^T \left(\frac{\partial b}{\partial \mathbf{p}} - \frac{\partial K}{\partial \mathbf{p}} u \right).$$

One adjoint solve with the same matrix is therefore required for every wrapped implicit linear solve, irrespective of the number of design variables. Within the custom adjoint rule, the converged state is retained and one additional system with the same matrix is solved.

The resulting sensitivities are shown in Fig. 2, in which the isolated point-load case is evaluated at iteration 60, corresponding to half of the 120 optimization epochs. For the element sensitivity dC/dE_e , negative values are obtained and concentrated along the principal load paths. The thickness derivative dC/dt is concentrated at the wall interfaces, since an interface indicator is obtained by differentiation of the occupancy sigmoid. The mask derivative is localized in regions in which the macro-topology remains largely undecided, while an influence of the phase derivative on the orientation of already present walls is observed. Considering the shared scale of the diverging fields, the modulus and mask channels exhibit raw sensitivities approximately one order of magnitude larger than the thickness channel at this state.

The relative contributions of the individual channels are additionally quantified through pooled gradient-norm shares. At initialization, shares of 2.8%, 3.5%, and 93.7% are obtained for the phase, thickness, and mask fields, respectively, whereas the corresponding values on the final lattice are 24.8%, 9.5%, and 65.8%. During early topology formation, the dominant contribution is therefore associated with the mask, while an increasing contribution of the phase channel is obtained after the macro-topology has largely settled. The associated forward geometry fields and reverse sensitivity flow are shown in Fig. 3. Through the volume-tangent projection shown in the final panel of Fig. 2, a 34% reduction of the thickness-gradient norm is obtained at the selected state. This value refers to the projection of the element-resolved physical thickness derivative dC/dt , whereas the percentages reported for Fig. 4 refer to the projection of the joint coarse-control-space (a, m) gradient used by the optimizer.

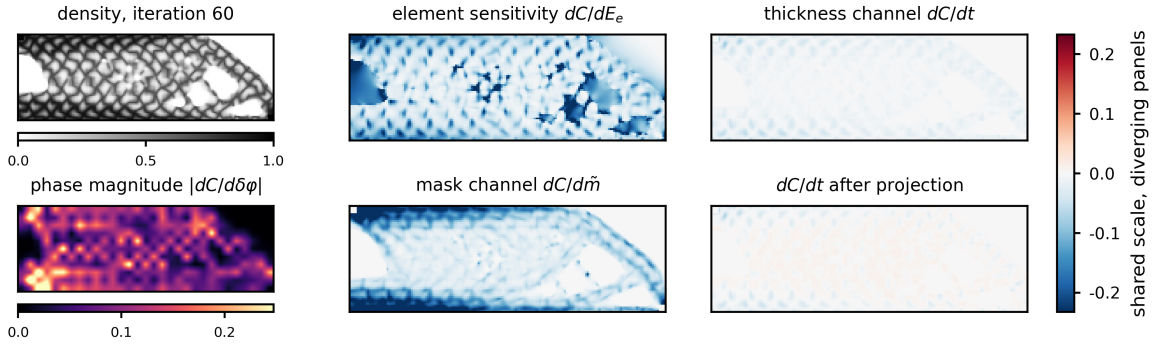


Figure 2: Overview of the element-resolved sensitivity fields for the unheated point-load case at iteration 60. The four diverging panels share one symmetric scale, while the phase panel shows the gradient magnitude on its own scale.

3.4 Finite element solution and volume constraints

The finite element calculations are conducted under plane-stress conditions by using bilinear quadrilaterals and a two-by-two Gauss rule. Specifically, one quadrature-generated reference stiffness matrix is used for all elements. Dirichlet conditions are imposed through a symmetric masking operator that preserves symmetry and positive definiteness on the free subspace. The stiffness action is evaluated matrix-free through gather, multiply, and scatter operations, while the resulting linear system is solved using Jacobi-preconditioned conjugate gradients. During optimization, the relative tolerance is linearly tightened from 10^{-4} to 10^{-6} over the continuation. All values reported in the present work are recalculated using a tolerance of 10^{-9} .

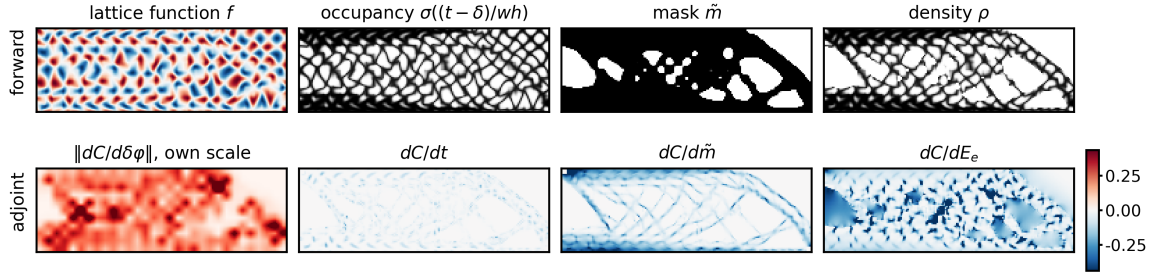


Figure 3: Overview of the forward fields and reverse sensitivities of the final reference lattice, including the distribution of the compliance gradient across the control channels.

The prescribed volume is satisfied at every iteration without an additional penalty to the objective. A scalar shift is applied to all thickness and mask controls before their bounded mappings and is bisected until the target volume is reached. This scalar restoration is placed outside the differentiated path. Given the nearly sign-uniform component of the compliance gradient, the componentwise normalization of Adam would otherwise lead to the conversion of a substantial part of the update into an approximately uniform dilation that would subsequently be removed by the volume restoration. The thickness and mask gradients are therefore first projected onto the tangent of the volume constraint, $g \leftarrow g - (n^T g / n^T n)n$, with $n = \partial V / \partial \mathbf{p}$ restricted to these two control blocks. Volume changes induced by the phase controls are subsequently absorbed by the scalar restoration. The prescribed volume is thereby restored exactly after every optimizer step. The differentiated path contains the volume-tangent projection of the thickness and mask gradients, while the scalar phase-induced restoration is applied subsequently outside this path.

The relation between the raw gradient, the volume-normal component, and the applied tangential update is illustrated for epochs 61 and 62 in Fig. 4. At these epochs, the volume-normal component has a norm equal to 98.8% and 99.0% of the raw gradient norm, respectively, while the corresponding tangential component has a norm equal to 15.7% and 14.3%.

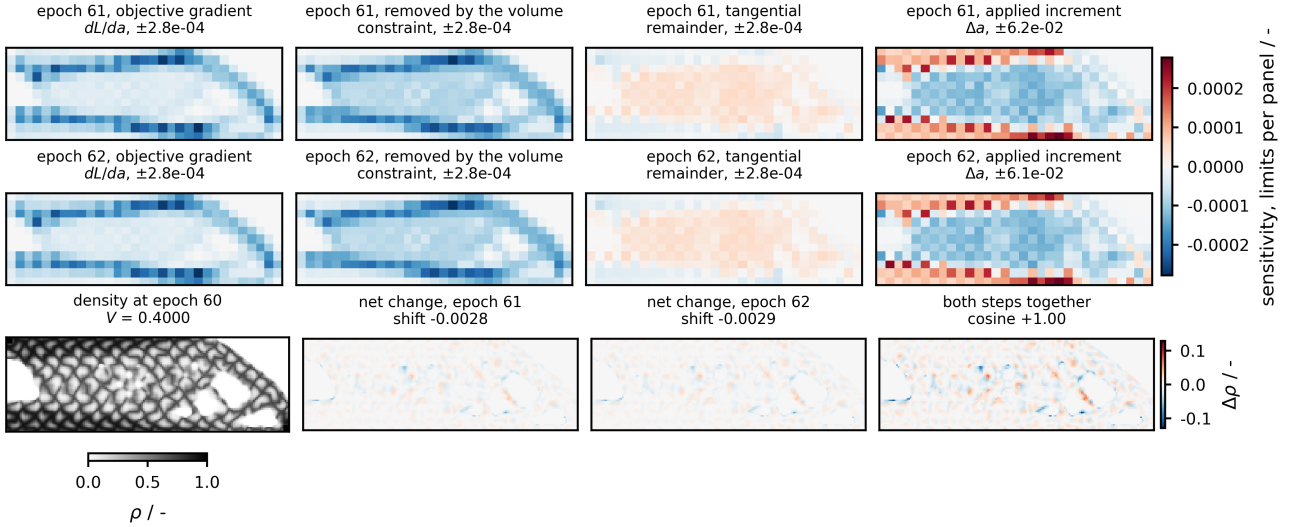


Figure 4: Exemplary overview of consecutive optimization steps based on the gradient, the volume-normal component, the tangential remainder, the applied increment, and the resulting density changes.

The optimization is divided into six stages of twenty iterations. Over the continuation, the interface half width is reduced from $2.0h$ to $0.35h$, the mask sharpness is increased from $\beta_m = 4$ to 128, the solver tolerance is tightened from 10^{-4} to 10^{-6} , and the step size is damped to one quarter of its initial value.

3.5 Environmental state couplings

In the present framework, geometry and the evolving material state are considered within one differentiable optimization problem. The geometry-dependent exposure $s = s(\mathcal{G}, u)$ is obtained from the synthesized geometric fields $\mathcal{G}$ and, where required, from the current displacement. Based on this exposure, the internal state z is determined from a residual $R_z(z; \mathcal{G}, u) = 0$, which may be formulated algebraically for a calibrated dose or incrementally for an evolving state. The resulting property field $E^* = E^*(\mathcal{G}, z)$ is used in the second equilibrium

$K(E^*)u^* = b(E^*)$, with the adjoint rule of Section 3.3 being applied to each implicit linear solve. The geometry, exposure, state evolution, local properties, and mechanics are thereby traversed within the same forward and reverse chain, while the optimizer, continuation, and volume treatment are retained across the individual models.

The examined cases differ in the origin and evolution of the state dependence. Design dependence is introduced through the local stiffness for the prescribed thermal excitation, through the mechanical solution for the stress-amplitude exposure, and through the surface and transport geometry for the chemical and tissue-deposition states, with transport and time stepping additionally included for the latter. Within the applied two-dimensional regime, the common coupling architecture is represented through the individual state laws.

Thermoelastic coupling

Under mechanical constraint, a prescribed temperature field generates internal forces proportional to the local stiffness, through which the applied excitation becomes design-dependent. The equivalent nodal force is $f_{\text{th}}(E) = \sum_e E_e \alpha \Delta T_e \int \mathbf{B}^T \hat{\mathbf{D}} [1, 1, 0]^T dV$, where the product $\alpha \Delta T_e$ is calibrated on the initial design. The heating ratio $\chi = f_{\text{th}}^T u_{\text{th}} / (b^T u_b)$ specifies the pure-thermal to pure-mechanical compliance ratio and is set to 0.5 or 1.5. The combined excitation is supplied to the wrapped equilibrium as a traced input, whereby the derivative contains both the stiffness operator and the design-dependent right-hand side.

Cyclic deterioration and thermal acceleration

The influence of repeated loading on the long-term material state is approximated by accumulating cyclic deterioration as a Palmgren-Miner sum of a Basquin S-N law [34], while continuous-time high-cycle fatigue alternatives have also been formulated [35]. The element amplitude is given by the von Mises value s_e of the stiffness-consistent centroid stress. The service life is traversed in n_b blocks, with the Miner contribution of block k being calculated according to the Basquin relation

$$\Delta \xi_e^{(k)} = \frac{a_e}{n_b} \left(\frac{s_e^{(k)}}{s_{\text{ref}}} \right)^{m_B}, \quad (2)$$

$$D_e = D_{\text{max}} \left[1 - \exp \left(- \sum_k \Delta \xi_e^{(k)} \right) \right], \quad (3)$$

with the Basquin slope $m_B = 9$, $n_b = 3$ blocks, $D_{\text{max}} = 0.8$, and the acceleration a_e given below. The equilibrium is re-solved on the partially deteriorated stiffness after every block, such that stress redistribution over the service life is included in the amplitude $s_e^{(k)}$. The modified modulus is $E_{\text{aged}} = E_{\text{min}} + (E - E_{\text{min}})(1 - D)$. Re-solving the equilibrium on this modulus, $K(E_{\text{aged}})u_a = b_{\text{mech}} + f_{\text{th}}(E_{\text{aged}})$, yields the end-of-life compliance $C_a = b^T u_a$. The virgin compliance $C_v = b^T u$, the relative aged-to-virgin compliance change $\Pi = (C_a - C_v)/C_v$, and the mean effective damage $\bar{D}$ are reported in addition. The reference stress is calibrated by bisection to a prescribed aged-to-virgin compliance ratio on the initial design and subsequently held fixed.

Thermal acceleration is calculated according to the Arrhenius relation. The normalized element temperature rise θ_e is mapped to the absolute element temperature $T_e = T_{\text{ref}} + \Delta T_{\text{max}} \theta_e$, and the rate ratio for every increment is given by

$$a_e = \exp \left[\frac{E_a}{R} \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T_e} \right) \right], \quad (4)$$

with the activation energy $E_a = 90$ kJ/mol, the gas constant R , the reference temperature $T_{\text{ref}} = 296.15$ K, and a surface rise $\Delta T_{\text{max}} = 12$ K, for which a peak acceleration of 4.15 is obtained. The same accelerated exposure is used during calibration, optimization, and every reported evaluation.

Photolytic and oxidative deterioration

Two further ageing mechanisms are considered in the mechanism-combination study. Photolytic exposure is prescribed through the top-surface profile

$$s_{\text{uv}} = \exp(-(H - y)/\ell_{\text{uv}}), \quad (5)$$

with $\ell_{\text{uv}} = 3$ element edges. Oxidative exposure is obtained from the concentration field

$$\nabla \cdot (\mathcal{D}_{\text{ox}} \nabla c_{\text{ox}}) = \frac{r_{\text{ox}} K_c}{K_c + c_{\text{ox}}} c_{\text{ox}}, \quad (6)$$

which is approximated by two fixed-point iterations. The diffusivity is high in the pore space and low in the solid, while $r_{\text{ox}} = \rho/\ell_{\text{ox}}^2$ is obtained from the geometric occupancy. The nominal reaction-length parameter is

$\ell_{\text{ox}} = 6$ element edges, with the realized penetration being governed by the local diffusivity and pore geometry. Unit concentration is imposed on the outer boundary of the computational domain by a weighted contribution to the operator and right-hand side. Differentiation is performed directly with respect to the continuous occupancy field.

Within the mechanism-combination study, the cyclic exposure is represented by $s_{\text{dy}} = s_e/s_{98}$, normalized by the initial 98th percentile, and every active exposure is converted into a common algebraic damage field according to $D_i = D_{\text{max}}[1 - \exp(-(\kappa_i s_i)^2)]$. One scalar severity metric is calibrated per condition to the same initial aged-to-virgin compliance ratio. For combined mechanisms, the multiplicative stiffness degradation is applied according to

$$E_{\text{aged}} = E_{\text{min}} + (E - E_{\text{min}}) \prod_i (1 - D_i). \quad (7)$$

Mechanical characteristics of tissue ingrowth

A stiffness-gaining state is examined through chondrocyte-mediated neotissue deposition. Tissue development within the scaffold pore space is evaluated under geometry- and flow-dependent conditions. The formulation of the individual model components is based on the perfusion-flow and shear-stress model of Guyot et al. (2015) [52], the coupled curvature- and shear-stimulated growth model of Guyot et al. (2016) [47], and the model-reduction and Bayesian optimization study of perfusion-culture conditions by Mehrian et al. (2018) [53]. The contribution of the neotissue to the local stiffness is represented by the parallel scaffold-tissue mixture introduced below. The transport, geometric, and mechanical stimuli are included in the exposure. ECM generation in an unvascularized construct is limited by a stationary balance between reaction and diffusion with saturating uptake in the pore phase. Static and perfused conditions are distinguished through the uptake parameter, while the effective nutrient penetration additionally depends on the local pore fraction. Geometric accessibility is represented by the clearance between the wall and a void element, consistent with curvature-dependent growth on scaffold surfaces [46]. Dynamic loading is represented by a simplified unimodal window of the strain amplitude, consistent with matrix-synthesis increases reported in chondrocyte-seeded constructs [48] and summarized in a critical review [49]. Under perfusion, a corresponding window of the Darcy wall-shear surrogate is added in accordance with coupled curvature- and shear-stimulated tissue growth [47].

The cell density is treated as an evolving state, since matrix deposition is permitted where cells are present. The cells spread by density-dependent diffusion, proliferate at a nutrient-limited rate, and die under insufficient nutrient supply. The present surrogate considers nutrient-mediated growth, while additional fields for degradation products can be integrated through the same differentiated path. Matrix synthesis is restricted according to nutrient access and the available pore space, while the mechanical stimuli are added to an unstimulated baseline. Based on the resulting matrix fraction, the modulus is modified through a parallel scaffold-tissue mixture and a swelling eigenstrain is introduced. The growth, transport, and mechanics are advanced over increments. The nutrient field is updated through a two-step fixed-point approximation, and the mechanical equilibrium is re-solved at each increment for conditions containing a mechanical stimulus.

The same differentiable state-coupling architecture is applied to deterioration and the stiffness-gaining tissue surrogate with structurally distinct evolution laws. In both cases, the extent of the state is controlled by a calibrated scalar, the modified modulus is included in the compliance objective, and the same continuation, optimizer specifications, volume restoration, and binary evaluation are applied. A resorbing-scaffold surrogate is additionally considered by coupling the local tissue deposition and a concentration-dependent scaffold-stiffness reduction to the common nutrient-access field, such that load transfer from the ageing scaffold to the steadily forming tissue is evaluated within one coupled chain.

3.6 Evaluation and reproducibility

After the prescribed continuation, all relaxed designs are evaluated at the nominal threshold and at a volume-matched threshold obtained by bisecting a uniform thickness offset. The resulting binary structures are subsequently re-solved, such that the difference between relaxed and binary performance is determined at the prescribed volume.

Environmental variations

The geometric representation, optimizer, continuation, and thresholding protocol are retained throughout the environmental campaigns. The load matrix combines a point load at the symmetry edge, a point load at half length, and a distributed load over the top edge with no heating and with the two calibrated heating ratios $\chi = 0.5$ and $\chi = 1.5$. An identical deterministic uniform initialization is used for all conditions. The prescribed temperature decreases exponentially from the top surface with a nominal depth of six elements. The load cases differ in their prescribed load-introduction regions, since a non-design strip is included along the loaded edge for the distributed condition and examined separately through a matched-strip control. For assessment of

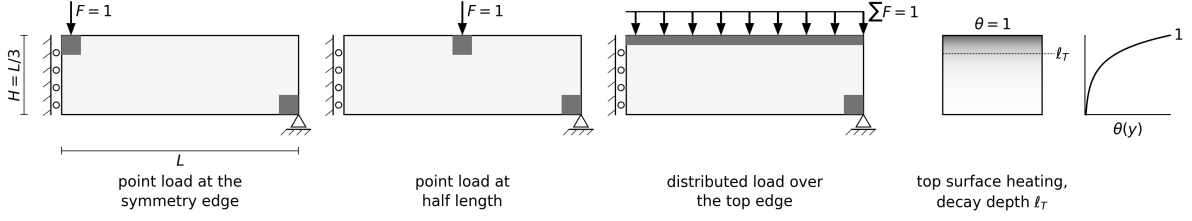


Figure 5: Overview of the mechanical load cases and the applied temperature field used in the condition matrix. Gray blocks denote non-design solid pads.

the response under deteriorating material properties, the designs under the same environmental conditions are optimized with respect to the final state under cyclic deterioration. For each condition, the severity is calibrated on the common initial design to yield an aged-to-virgin compliance increase of 50%, after which the end-of-life compliance is used as the objective. The influence of the exposure pattern is examined using the unheated symmetry-edge point-load case as the reference condition. Oxidation is represented by the solved oxygen field as an approximation of the locally present oxidation concentration, photolytic ageing by a prescribed top-surface layer resembling exogenous UV exposure, and cyclic deterioration by the stress-amplitude field. The equal-initial-harm comparison comprises the reference condition, the three individual ageing mechanisms, their three pairwise combinations, and the triple combination. For every active set, one severity scalar is calibrated on the initial design to the same 50% aged-to-virgin compliance increase, through which the spatial exposure patterns are compared at equal initial harm. In a second evaluation, the single-mechanism severities are retained in the combinations so that interactions between the ageing mechanisms are assessed without redistribution of the calibrated dose. The stationary-amplitude algebraic law is used in this campaign to compare the spatial exposure patterns, while the three-block recurrence including mechanical stress redistribution is used in the ageing matrix.

On the fixed geometry, the pairwise interaction contrast is defined from the underlying compliance values as

$$I_{AB} = C_{AB} - C_A - C_B + C_0.$$

The complementary third-order interaction contrast follows as

$$I_{ABC} = C_{ABC} - C_{AB} - C_{AC} - C_{BC} + C_A + C_B + C_C - C_0.$$

Positive values indicate a super-additive compliance response relative to the corresponding additive decomposition on the fixed geometry. The contrasts therefore quantify the combined response of the selected multiplicative stiffness law and the global compliance functional.

For attribution of the design sensitivity, the total projected gradient $\mathbf{g}_{\text{tot}}$ is decomposed into the frozen-state contribution $\mathbf{g}_{\text{frz}}$ and the state-mediated contribution

$$\mathbf{g}_z = \mathbf{g}_{\text{tot}} - \mathbf{g}_{\text{frz}}.$$

The relative state-mediated magnitude and the rotation of the projected design-gradient direction are defined as

$$r_z = \frac{\|\mathbf{g}_z\|_2}{\|\mathbf{g}_{\text{tot}}\|_2}, \quad \vartheta_z = \arccos\left(\frac{\mathbf{g}_{\text{tot}}^T \mathbf{g}_{\text{frz}}}{\|\mathbf{g}_{\text{tot}}\|_2 \|\mathbf{g}_{\text{frz}}\|_2}\right).$$

The mechanical placement factor is defined as $P_z = \langle z \rangle_{\text{mech}} / \langle z \rangle_{\text{ref}}$. For deterioration and resorption, the mechanical average is weighted by the element contribution to the virgin, non-aged compliance and the reference average by the material occupancy. For neotissue as a stiffness-gaining state, the effective tissue fraction is compared through mechanically weighted and unweighted pore-space averages. A value of $P_z > 1$ therefore indicates concentration of the state in mechanically relevant regions. The norms and angles are evaluated after the same volume-tangent projection used by the optimizer and in the Euclidean metric of the adopted control vector $\mathbf{p}$. They consequently quantify sensitivity attribution within the present parameterization and vary under rescaling or reparameterization of the control variables. Since r_z is a norm ratio, values above unity may arise through partial cancellation between the frozen-state and state-mediated contributions.

The stiffness-gaining tissue coupling is additionally evaluated on the final unheated point-load geometry under static and perfused culture, with both conditions being considered with and without dynamic loading. The fixed-geometry comparison isolates the culture environment, after which one geometry per condition is obtained through tissue-state-aware optimization. For a standardized comparison, all reported states are calibrated to the same tissue yield of 10% of the domain. A neotissue-to-scaffold modulus ratio of $\eta = 0.1$ is applied throughout the composite optimizations. The underlying numerical settings of the lattice representation and continuation are summarized in Table 1, while the additional parameters of the tissue-deposition approximation are listed in Appendix D.

Binary robustness benchmark

For assessment of the extracted deterioration-aware design, a further benchmark is conducted for the unheated symmetry-edge point-load case based on the penalized lattice-wall stiffness exponent $q_{\text{lat}} = 3$. The deterioration reference parameter $s_{\text{ref}} = 0.226694$ is retained across all branches. Three blind reference geometries are optimized on a grid of 180×60 elements from reproducible initializations. For every initialization, a deterioration-aware continuation is then initialized from the corresponding converged blind geometry with reset Adam moments and 40 iterations at the final continuation state. The extracted blind and aware geometries are volume-matched to a material fraction of 0.40 and evaluated at 180×60 , 360×120 , and 720×240 . For the deterministic pair, the full derivative through the deterioration state is compared with a frozen-state derivative under an otherwise identical continuation. The temporal discretization is assessed on the final aware geometry based on 3, 6, 12, and 24 deterioration sequences.

Table 1: Overview of the core numerical settings of the representation and the continuation. Lengths correspond to nondimensional element edges.

Symbol	Meaning	Value
k_0	Nominal wave number	$2\pi/12$
$t_{\min}, t_{\max}$	Half-thickness bounds	0.8, 2.6
Δ_c	Control-grid spacing	6
$\det \mathbf{J}/k_0^2$	Determinant target	≥ 0.25
$\text{cond } \mathbf{J}$	Anisotropy target	≤ 4
$N_{\min}$	Wave-number guard	8
$w_{\text{det}}, w_{\text{band}}, w_{\text{cond}}$	Penalty weights	10, 10, 1
w	Interface width schedule	$2.0 \rightarrow 0.35$
β_m	Mask sharpness schedule	$4 \rightarrow 128$
ϵ_{∇}	Gradient-norm floor	0.05
ϵ_f	Regularization of $ f $	10^{-3}
m_B	Basquin slope	9
n_b	Miner blocks	3
E_a	Activation energy	90 kJ/mol
$\Delta T_{\max}$	Maximum surface temperature increase	12 K

4 Results and discussion

4.1 Response under mechanical and thermal loads

The applied mechanical load cases and prescribed thermal field are shown in Fig. 5. As summarized in Table 2 and Fig. 6, the expected compliance ordering is obtained for the mechanical conditions. The highest compliance is obtained for the symmetry-edge point load, consistent with its longer active lever arm. When the point load is shifted to $x = L/2$, both the lever arm and the compliance are reduced, while a comparable reduction is obtained for the distributed load.

Under the applied thermal loads, the total compliance increases for every factorial combination, while the magnitude of the increase is governed by the relation between load introduction and the heated boundary layer. For the localized point loads, contact between the load-bearing structure and the thermally loaded layer is restricted to the load-introduction point, through which the compliance increase remains between 4% and 41%. The distributed load is introduced along the complete heated edge, so that the load-bearing structure remains coupled to the constrained expansion. The thermal load-vector work share consequently approaches one half at $\chi = 0.5$ and reaches 76% at $\chi = 1.5$, where a reorganization around the thermally induced stress is obtained as shown in Fig. 7. A matched control, in which the same two-element solid strip is imposed on the symmetry-edge point-load case, reaches the same thermal share of 76% and a total compliance of 684.9. The response is therefore likely governed to a substantial extent by the complete load-introduction geometry, including its non-design regions, since the force distribution alone does not account for the observed variability.

The evolution of the unheated point-load design is shown in Fig. 8. The macro-topology is established primarily during the first two continuation stages, after which the wall interfaces are sharpened and the mask is consolidated while the principal members remain largely unchanged. The compliance reaches 242 at iteration 60 and increases to 251 in the final state, corresponding to an increase of approximately 3% during late-stage interface sharpening.

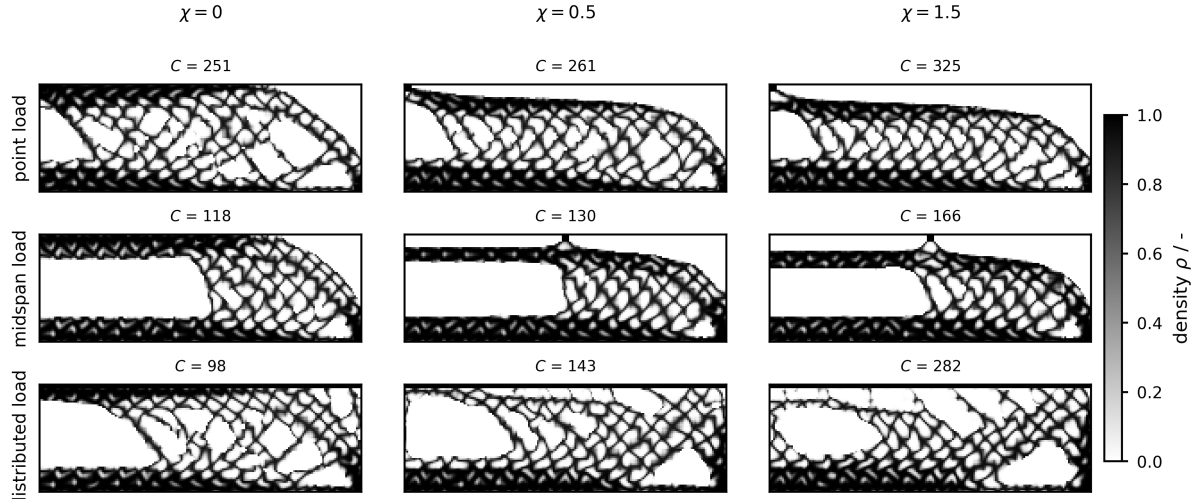


Figure 6: Overview of the final structures of the load-thermal matrix. Rows denote load cases, columns the heating ratio χ , and panel titles the total compliance C .

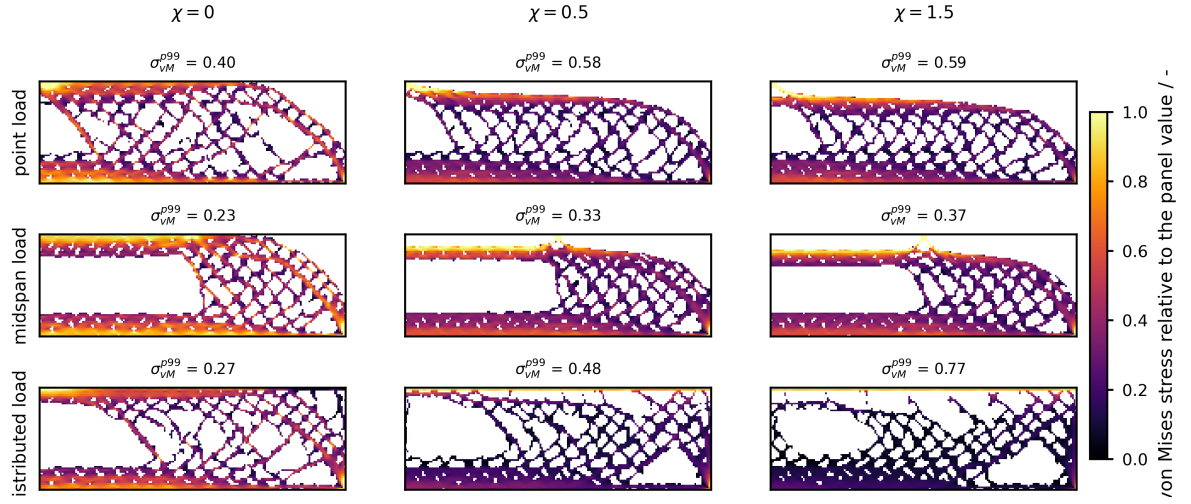


Figure 7: Overview of the thermomechanical von Mises fields of the nine matrix cells. Each panel is scaled to its own 99th percentile over the solid elements.

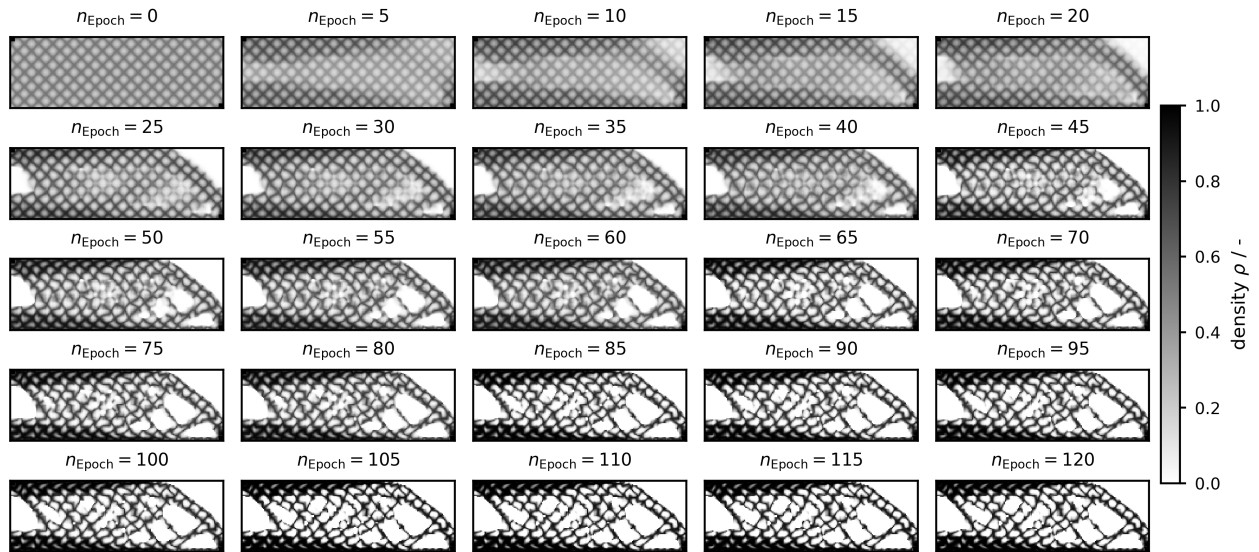


Figure 8: Overview of the evolution of the unheated point-load design from initialization to epoch 120, with every fifth optimization epoch shown.

Table 2: Overview of the total compliance of the full-factorial matrix at volume 0.40 after 120 epochs. Thermal-work fractions $u^T f_{\text{th}}/C$ are given in parentheses.

Load case	$\chi = 0$	$\chi = 0.5$	$\chi = 1.5$
Point load, symmetry edge	250.6	261.1 (18%)	324.8 (29%)
Point load, half length	117.9	130.0 (15%)	166.0 (33%)
Distributed top load	98.5	143.1 (46%)	281.7 (76%)

4.2 Optimization under deteriorating material properties

The numerical results are summarized in Table 3, while the corresponding final geometries and the damage distribution of the unheated symmetry-edge point-load case are shown in Figs. 9 and 10. Because the deterioration law is driven by the stress amplitude, the resulting state is concentrated along the principal load paths and reaches the prescribed saturation in the most highly stressed walls. Across the aged matrix, increasing mean damage is associated with increasing local stress concentration, and the localization is intensified as mechanically efficient members emerge during optimization. The unheated point-load design consequently attains a low aged compliance while exhibiting a larger aged-to-virgin compliance increase than the uniform lattice on which the severity was calibrated. With an increasing thermal share, the relative compliance change Π decreases and becomes negative at $\chi = 1.5$ for the two point-load cases. Under the adopted boundary conditions, deterioration reduces both the stiffness and the stiffness-proportional thermal excitation, through which a negative compliance change can be obtained in the linear thermoelastic model. The distributed condition remains positive over the examined range.

When the deterioration-blind designs of Table 2 are evaluated under identical ageing within the linearly interpolated formulation, blind-to-aware end-of-life compliance ratios from 1.22 to 1.72 are obtained. The largest ratio is obtained at the intermediate thermal share, where deterioration is already accelerated by the prescribed temperature field while the blind design remains predominantly governed by the mechanical objective. Under strong heating, both designs are reorganized by the thermal excitation, whereas ratios from 1.22 to 1.33 are obtained under ambient thermal conditions.

Table 3: Overview of the aged matrix, including the end-of-life compliance C_a , ageing penalty Π , mean effective damage $\bar{D}$, aged compliance C_a^{blind} of the deterioration-blind designs, and gain $G = C_a^{\text{blind}}/C_a$.

Cell	C_a	Π	$\bar{D}$	C_a^{blind}	G
Point, symmetry edge, none	548.4	+0.64	0.120	730.5	1.33
Point, symmetry edge, 0.5	543.4	+0.26	0.141	911.3	1.68
Point, symmetry edge, 1.5	752.7	-0.16	0.203	1141.6	1.52
Point, half length, none	331.8	+0.84	0.184	439.9	1.33
Point, half length, 0.5	324.3	+0.14	0.214	556.1	1.72
Point, half length, 1.5	479.6	-0.30	0.286	634.9	1.32
Distributed, none	155.2	+0.46	0.052	189.3	1.22
Distributed, 0.5	237.8	+0.35	0.153	400.7	1.69
Distributed, 1.5	430.3	+0.09	0.295	590.7	1.37

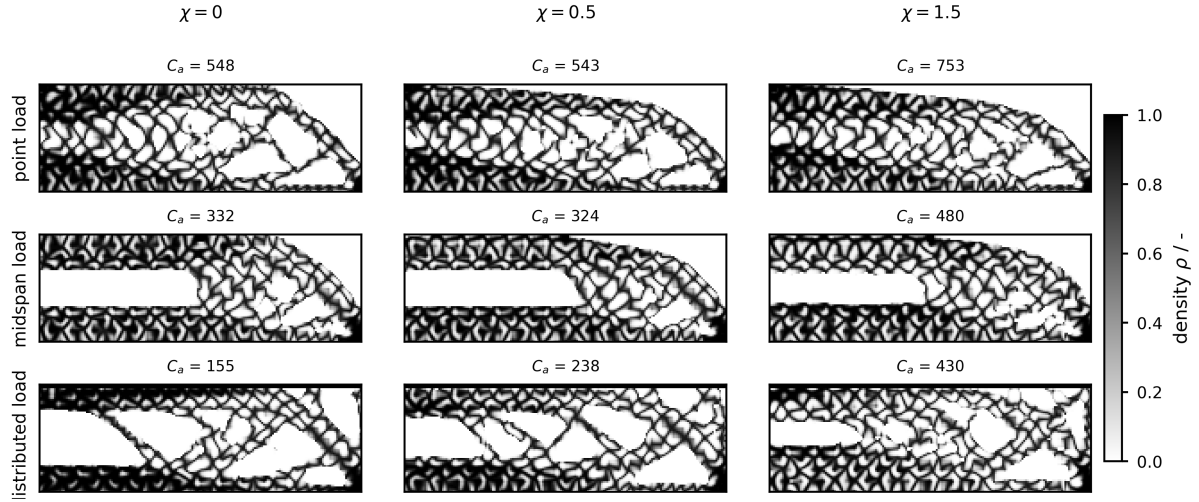


Figure 9: Overview of the deterioration-aware final designs of the aged matrix. Rows denote load cases, columns the heating ratios, and panel titles the aged compliance.

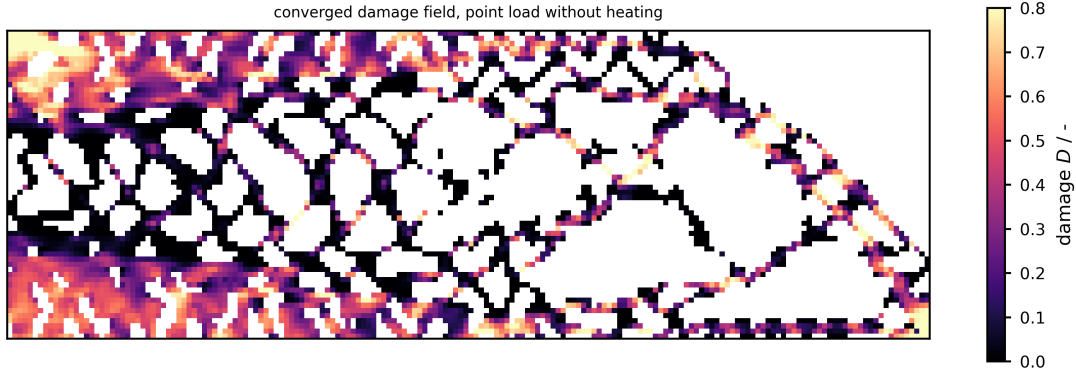


Figure 10: Overview of the cyclic damage field of the final unheated point-load design.

4.3 Optimization responses to applied exposure pattern

As shown in Table 4 and Fig. 11, the mechanisms can be ordered according to the extent to which the load paths can be displaced away from the corresponding exposure. The oxygen field is supplied at the outer domain boundary and transported through the pore phase, so that its concentration within the load-bearing walls depends on the wall thickness and pore connectivity of the current design. Oxidation consequently yields the lowest aged compliance. Under the prescribed ultraviolet exposure, which decreases exponentially from the top surface, material placement is restricted within a fixed near-surface region and an intermediate aged compliance is obtained. Cyclic deterioration remains intrinsically coupled to the load paths, through which every structurally required wall is exposed and the highest aged compliance is obtained.

For the equal-initial-harm combinations, the response is governed by the least avoidable active exposure and moderated by the distribution of the calibrated severity among the active channels. The combination of UV radiation and cyclic loading remains close to the cyclic result, while the oxidation-containing cyclic combinations yield lower compliances, consistent with a reduced relative contribution of the stress-driven channel when the common severity is shared. Interestingly, the combined ageing under the influence of UV irradiation and oxidation remains close to the oxidation result. The calibrated severity of the thin UV layer exceeds those of oxidation and cyclic deterioration by more than one order of magnitude, which is consistent with the spatially restricted and superficial character of the prescribed ultraviolet exposure.

The binary evaluation is additionally affected by the geometry dependence of the exposure. Direct application of the thresholded wall geometry increases the local stresses, through which both the stress amplitude and the subsequent cyclic deterioration increase. The binary aged compliances of the stress-coupled conditions consequently exceed those of the surface-controlled mechanisms by several times.

On a fixed geometry, the triple combination reaches an increase of a factor of 1.38 over the additive prediction under consideration of the multiplicative reduction alongside 13.9 times the corresponding prediction under an additive reduction, as summarized in Table 5. The principal super-additive response is therefore generated by the constitutive combination rule. Pairwise interaction contrasts of +19 and +4 are obtained for the oxidation-

containing pairs and a contrast of +99 for the combined ultraviolet and cyclic loading, where common load-bearing regions are affected by both fields, while the third-order contrast of the triple combination is +12. Relative to the fixed-geometry evaluations, the re-optimized compliances range from a 6.8% decrease to a 24.5% increase, while a reduction was obtained only for ageing under combined UV irradiation and oxidation. The overall magnitude is governed jointly by the combination law and the spatial overlap of the exposures. However, the cyclic single-mechanism result remains close to the corresponding cell of the deterioration matrix, with compliances of 529.4 for the stationary-amplitude factorial law and 548.4 for the blockwise state evaluation.

The response over the normalized exposure dose is shown in Fig. 12. With the onset of the applied damage laws, the mean solid stiffness decreases and the compliance increases. An early saturation is obtained for the UV-irradiated layer, while the cyclic exposure remains strongly localized as indicated by the exceedance curves. The three spatial signatures are distinguished in the mid-span profiles as a prescribed surface layer, a transport-controlled oxygen field, and a load-path deterioration field. The evolution of the triple-mechanism optimization is shown in Fig. 13. As in the reference load case, the macroscopic topology is established early, while the subsequent interface sharpening produces a larger compliance increase. The aged compliance rises from a value of 408.3 at iteration 60 to 446.1 in the final state, which exceeds the minimum obtained with the softer interface by approximately 14%. This behavior is consistent with the concentration of the stress amplitude in sharp, thin walls.

The contribution of the differentiated state path is quantified in Fig. 14 and Table 6. For the prescribed UV profile, the state-mediated gradient vanishes as expected for a geometry-independent exposure field. On the final reference lattice, oxidation yields $r_z = 0.558$ and a rotation of the projected design-gradient direction by 33.9° , while cyclic deterioration yields $r_z = 0.845$ and a rotation of 56.3° . On the common initial lattice, the corresponding values are $r_z = 0.089$ and 2.5° for oxidation and $r_z = 1.008$ and 74.7° for cyclic deterioration. Differentiation through the ageing state therefore changes both the magnitude and the direction of the design sensitivity and, consequently, the geometric optimization path.

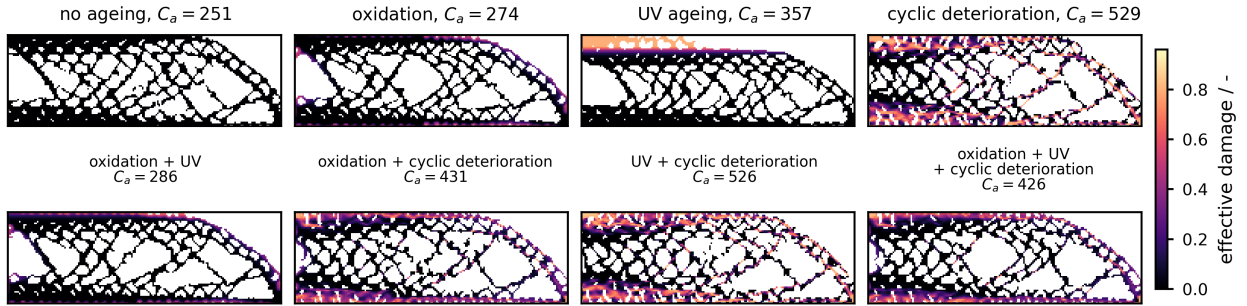


Figure 11: Overview of the damage fields of the mechanism-combination campaign at equal initial harm.

Table 4: Overview of the mechanism-combination campaign at equal initial harm, including the end-of-life compliance C_a , ageing penalty Π , mean effective damage $\bar{D}$, calibrated severity κ , and binary aged compliance.

Cell	C_a	Π	$\bar{D}$	κ	Binary C_a
No ageing	250.6	0.00	0	0	333.7
Oxidation	273.6	0.09	0.101	1.16	342.2
UV ageing	357.3	0.27	0.141	25.4	466.4
Cyclic deterioration	529.4	0.61	0.108	1.09	6174.6
Oxidation + UV	285.9	0.13	0.109	1.09	382.4
Oxidation + cyclic deterioration	431.5	0.51	0.141	0.89	3078.5
UV + cyclic deterioration	526.5	0.65	0.114	1.05	5519.1
Oxidation + UV + cyclic deterioration	425.7	0.52	0.138	0.86	2726.8

Table 5: Overview of the mechanism combinations at the single-mechanism severities, comprising the reoptimized compliance C_a^{opt} , fixed-design compliance C_a^{fixed} , and interaction contrast I_{AB} or I_{ABC} .

Combination	C_a^{opt}	C_a^{fixed}	I_{AB} / I_{ABC}
Oxidation + UV	465.5	499.2	+19
Oxidation + cyclic deterioration	503.7	404.5	+4
UV + cyclic deterioration	731.0	672.9	+99
Ox. + UV + cyclic deterioration	754.6	736.2	+12

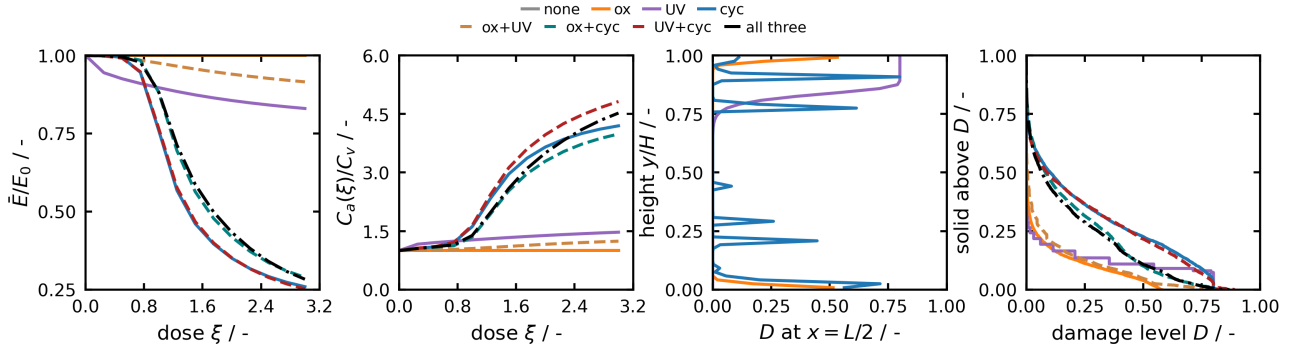


Figure 12: Overview of the dose-resolved response of the eight mechanism-combination cells, comprising the mean solid stiffness, compliance, mid-span height profiles, and damage exceedance curves.

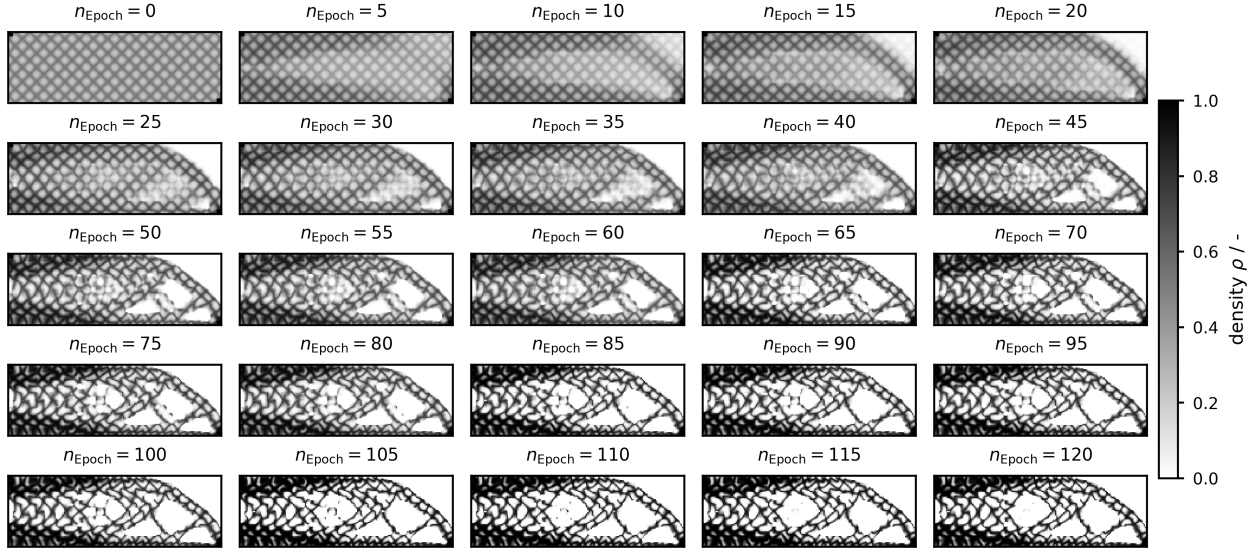


Figure 13: Overview of the evolution of the point-load design under the combined effects of oxidation, UV exposure, and cyclic deterioration from epoch 0 to 120.

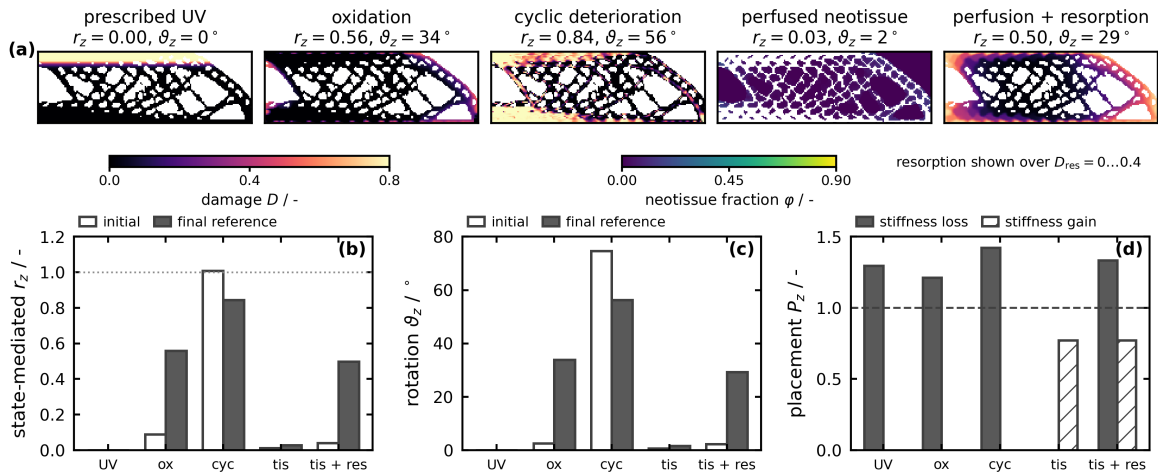


Figure 14: Overview of the state-mediated sensitivities on the deterioration-blind reference lattice, including the representative state fields, relative state-mediated magnitude r_z , gradient rotation θ_z , and mechanical placement factor P_z .

Table 6: Overview of the state-mediated sensitivity on the common projected initial lattice and the final deterioration-blind reference lattice, including the relative magnitude r_z , gradient rotation ϑ_z , mechanical placement factor P_z , and phase share of the state-mediated block norm.

Condition	$r_z^{(0)}$	$r_z^{(f)}$	$\vartheta_z^{(f)}$	$P_z^{(f)}$	ph.
Applied UV	0.000	0.000	0.0°	1.29	0.00
Oxidation	0.089	0.558	33.9°	1.21	0.76
Cyclic deterior.	1.008	0.845	56.3°	1.42	0.48
Static neotissue	0.015	0.018	1.0°	0.59	0.73
Perfused neotissue	0.012	0.028	1.5°	0.77	0.39
Static tis., resorp.	0.028	0.411	24.2°	0.59, 1.80	0.95
Perfused tis., resorp.	0.041	0.498	29.3°	0.77, 1.33	0.68

4.4 Optimization under approximated tissue deposition

As shown in Table 7 and Figs. 15 - 16, the spatial extent of tissue deposition is governed predominantly by transport. Static culture produces a concentrated boundary rim, whereas perfusion permits a more extensive distribution of tissue within the scaffold. Because the composite stiffness increases superlinearly with the local tissue fraction, the concentrated static deposit provides the larger mechanical contribution. Across the four non-resorbing culture conditions, tissue deposition can be shown to reduce the composite compliance and thus yields tissue gains in a range from 1.03 to 1.04 on the fixed point-load geometry.

The tissue-state-aware geometries are summarized in Table 8 and Fig. 17. The observed cultured response is, given the approximated description of tissue ingrowth we applied in this paper, determined by the transport, wall clearance, and strain window. For the four non-resorbing conditions, the design gains remain between 0.996 and 1.000, indicating only minor changes relative to the tissue-state-blind reference at the applied (neo-)tissue stiffness. The strongest geometry-specific adaptation is obtained for the perfused resorbing scaffold, for which a design gain of 1.036 is reached because tissue deposition and scaffold resorption are both coupled to transport access. The culture extent required for equal yield was found to be approx. 50 units under static culture and approx. 5 units under perfusion. Joint optimization of stiffness-promoting tissue deposition and stiffness-losing scaffold resorption is thereby demonstrated within one geometry-dependent differentiable state problem, in which both processes are coupled to the common concentration field and the resulting transfer of load from the scaffold to the tissue is resolved.

Table 7: Overview of the chondrocyte-mediated tissue conditions at equal tissue yield, including the extent Λ , tissue-weighted mean distance $\bar{\ell}$, composite compliance C_g , and tissue gain $G_{\text{tissue}} = C/C_g$.

Condition	Λ	$\bar{\ell}$	C_g	G_{tissue}
Static culture	49.4	9.5	240.8	1.04
Static culture with dynamic loading	52.4	9.5	240.7	1.04
Perfused culture	5.19	13.6	243.9	1.03
Perfused culture with dynamic loading	5.00	13.5	243.4	1.03

Table 8: Overview of the tissue-state-aware optimization for the four stable and two resorbing culture conditions, including the composite compliance C_g , stable-scaffold compliance C_s , tissue-state-blind compliance C_g^{blind} , design gain G_{design} , and equal-yield extent Λ .

Condition	C_g	C_s	C_g^{blind}	G_{design}	Λ
Static culture	241.5	252.0	240.8	0.997	49.5
Static culture with dynamic loading	240.9	251.3	240.6	0.999	60.3
Perfused culture	244.9	252.1	243.9	0.996	5.05
Perfused culture with dynamic loading	243.2	250.9	243.2	1.000	5.10
Static culture, resorbing scaffold	260.9	251.2	259.6	0.995	36.9
Perfused culture, resorbing scaffold	283.8	251.9	294.1	1.036	6.44

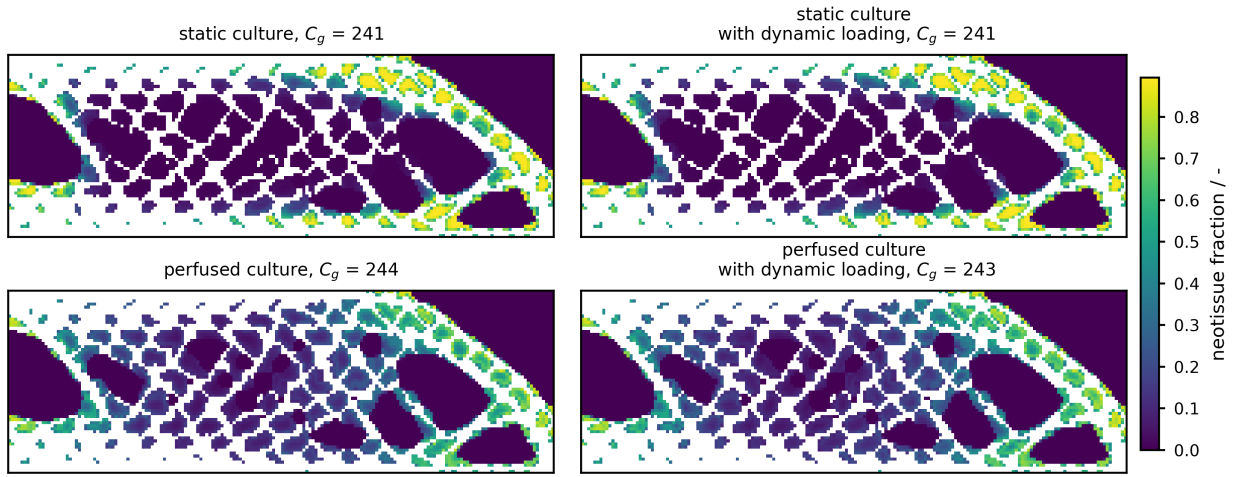


Figure 15: Overview of the (neo-)tissue distribution of the four culture conditions on the final point-load design at equal tissue yield.

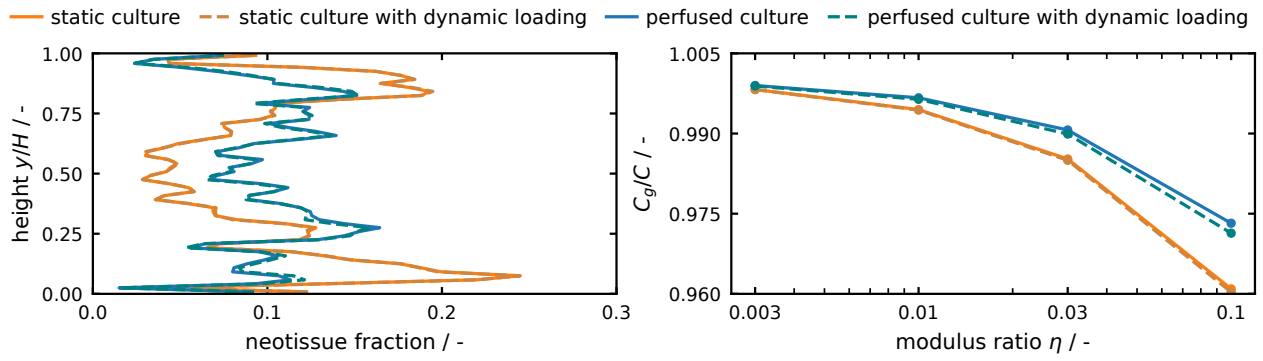


Figure 16: Overview of the height-resolved neotissue fraction under the four culture conditions and the sensitivity of the composite compliance to the neotissue-to-scaffold modulus ratio.

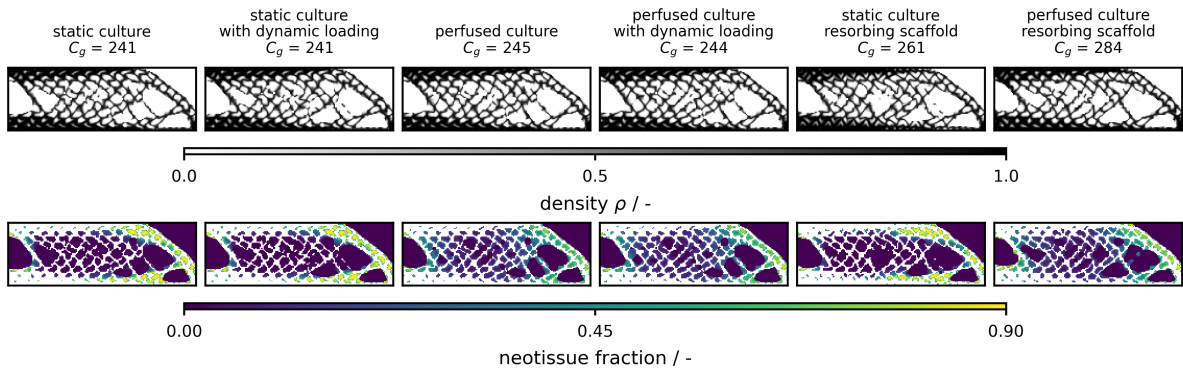


Figure 17: Overview of the tissue-state-aware final designs and corresponding neotissue fields for the four stable and two resorbing culture conditions.

4.5 Binary robustness of the extracted deterioration-aware design

The paired penalized-interface results are summarized in Table 9. At a resolution of 720×240 , the aged compliance of the volume-matched binary lattice is reduced by 7.72%, 7.87%, and 8.47% for the three paired initializations, respectively. The median reduction is 7.87%. Across all three evaluation resolutions, reductions from 7.24% to 9.10% are obtained. The deterioration-aware continuation therefore reduces the aged compliance of the extracted geometry.

Table 9: Overview of the paired penalized-interface benchmark for the unheated symmetry-edge point-load case. The blind and deterioration-aware geometries are optimized on the 180×60 grid with $q_{\text{lat}} = 3$ and evaluated as volume-matched binary structures at the stated resolutions.

Init.	Red. 180×60	Red. 360×120	$C_a^{\text{blind}} 720 \times 240$	$C_a^{\text{aware}} 720 \times 240$	Red. 720×240
0	9.04%	7.24%	1223.739	1129.300	7.72%
1	7.92%	8.73%	1262.990	1163.655	7.87%
2	8.90%	9.10%	1232.618	1128.182	8.47%

For the deterministic pair, the full derivative through the evolving deterioration state reduces the relaxed aged objective from 990.2 to 970.3, corresponding to 2.01%. After binary extraction, the full-state objective differs from the frozen-state objective by -0.48% , $+1.16\%$, and $+0.86\%$ at 180×60 , 360×120 , and 720×240 , respectively. The binary difference associated with the state-mediated derivative therefore remains within 1.2% across the examined resolutions.

At the 180×60 evaluation, the block-count study on the deterministic aware binary geometry yields aged compliances of 1074.8, 1052.0, 1030.6, and 1017.3 for 3, 6, 12, and 24 deterioration blocks. Increasing the number of blocks from 3 to 24 changes the aged compliance by 5.34%, while the 12- and 24-block values differ by 1.29%.

4.6 Relation to prior art and scope

The implicit lattice remains the design and evaluation geometry throughout optimization, with the integrable phase potentials retaining a coherent local orientation, the thickness field controlling the resolved walls, and the mask providing the macroscopic topology. A separate dehomogenization or lattice-reconstruction stage is thereby avoided. The background-grid projection permits the resulting geometry to be evaluated through resolved mechanics, while one adjoint solve is retained for every implicit linear state. For the reference cases, the lattice and feature-matched element-density descriptions yield relaxed compliances within two percent under the same model and optimizer.

The central extension beyond the initial mechanical design consists of the geometry-dependent evolving material state. The wall distance, material within the prescribed heated layer, stress amplitude, and pore connectivity are obtained from the current geometry. From these fields, the exposure and state evolution are determined, after which the state-dependent mechanical properties enter the objective and the derivative is returned through the complete chain. The integrable phase-potential warping, independent wall-thickness control, resolved background-grid mechanics, and evolving physical, chemical, or biological state fields are thus combined within one differentiable representation. Additional differentiable mechanisms can be integrated through corresponding exposure and state-evolution laws without modification of the underlying geometric and adjoint framework, while the warping, stretching, and rotation fields can be extended to broaden the accessible class of locally modified implicit lattice geometries.

5 Conclusion and outlook

In the present paper, direct gradient-based topology optimization was formulated for differentiable implicit lattice geometries whose environmental exposure and local material state depend on the geometry itself. Integrable phase potentials, a bounded wall-thickness field, and a macroscopic topology mask were coupled to resolved background-grid mechanics. Reverse differentiation through the exposure, state evolution, local properties, and mechanical response subsequently permitted optimization with respect to the aged or stiffness-modified state. Mechanism-specific geometric adaptations and substantial state-mediated changes of the design sensitivity were obtained across the thermal, cyclic, oxidative, photolytic, tissue-deposition, and resorption examples.

Within the penalized-interface benchmark, deterioration-aware continuations from three blind reference geometries reduced the aged compliance of volume-matched binary lattices by 7.72 - 8.47% at 720×240 . For the deterministic pair, inclusion of the state-mediated derivative reduced the relaxed aged objective by 2.01%. Direct lattice-geometry optimization differentiated through an evolving material state is thereby demonstrated together with a binary performance benefit.

Further development is directed toward three-dimensional lattice descriptions, experimentally calibrated exposure and state-evolution laws, stronger enforcement of geometric admissibility, and refined spatial and temporal discretizations. These extensions could easily be introduced within the proposed approach without changing the underlying optimization principle.

A Reference density formulation and mesh-resolution sensitivity

The lattice representation is compared with an element-density formulation using the same finite element model, loads, non-design regions, optimizer, volume restoration, continuation, and binary thresholds. A cubic stiffness interpolation is applied to the element-density field, while the lattice uses a cubically penalized mask and, in the principal campaigns, a linearly interpolated wall interface. The descriptions therefore differ in both their geometric restriction and their treatment of intermediate material. The density feature size is prescribed through a cone filter. A radius of half the lattice period is used to match the resolvable feature content, and a radius of 1.5 elements is included for the point-load case as the finest setting admitted by the mesh.

The relaxed and volume-matched binary objectives are summarized in Table 10. Comparable relaxed compliances are obtained for the two descriptions, with values of 250.6 for the lattice and 252.3 for the feature-matched density design under the point load and values of 98.5 and 99.5, respectively, under the distributed load. These differences contain both the geometric restrictions and the distinct interpolation laws. After thresholding, the density designs retain their compliance to within one percent, whereas losses of 53% under the point load and 146% under the distributed load are obtained for the linearly interpolated lattice. Because the diffuse interface occupies a fixed fraction of the twelve-element lattice period, binary extraction removes a systematic share of every wall. With a penalized lattice interface, the point-load binary compliance becomes 7% lower than the relaxed value, while the extraction gap of the strongly heated distributed condition is reduced to 5%.

Table 10: Overview of the element-density reference against the lattice on the same model and optimizer, for two load cases without heating.

Description	Relaxed	Thresh.	Loss, relative change
Point load at the symmetry edge			
Lattice	250.6	382.3	+53 %
Density, $r_f = 6$	252.3	249.7	-1 %
Density, $r_f = 1.5$	247.1	247.1	0 %
Lattice, $q_{\text{lat}} = 3$	301.3	278.9	-7 %
Distributed load			
Lattice	98.5	242.1	+146 %
Density, $r_f = 6$	99.5	99.1	0 %
Distributed load, $\chi = 1.5$			
Lattice	281.7	702.6	+149 %
Lattice, $q_{\text{lat}} = 3$	755.6	796.7	+5 %

Mesh sensitivity is additionally quantified for two conditions at an increased resolution of 360×120 elements with $h = 0.5$. The physical lattice period, thickness bounds, control spacing, load-introduction strip, thermal decay depth, and wave-number guard are retained from the base discretization. As summarized in Table 11, the relaxed compliance of the unheated point-load case changes by 3.6%, while the binary gap decreases from 53% to 17%. Under strong heating with distributed loading, the thermal load-vector work share remains between 76% and 78%, while the binary gap decreases from 149% to 16%. A moderate residual mesh sensitivity is therefore obtained for the relaxed objective, whereas the binary gap remains strongly dependent on the resolution of the diffuse interface. Increased resolution is particularly indicated where local stress concentrations enter the exposure or state-evolution law.

Table 11: Overview of the mesh-dependent relaxed and volume-matched binary compliances and the corresponding thresholding loss.

Cell	Grid	Relaxed	Thresh.	Loss
Point load	180×60	250.6	382.3	+53 %
Point load	360×120	259.7	304.3	+17 %
Distr., $\chi = 1.5$	180×60	281.7	702.6	+149 %
Distr., $\chi = 1.5$	360×120	317.8	367.1	+16 %

B Initialization sensitivity of the optimizer

The sensitivity to the initialization is assessed for the unheated symmetry-edge point-load reference. Four additional runs were initialized from control fields perturbed by reproducible normal noise with an amplitude of 0.05 in all three channels. As summarized in Table 12, the resulting compliance ranges from 241.2 to 245.7, compared with 250.6 for the deterministic initialization. The perturbed results are therefore 2% to 4% lower than the baseline.

Table 12: Overview of the initialization sensitivity of the reference cell, with Δ denoting the deviation from the deterministic baseline.

Initialization	C	Δ
Unperturbed	250.6	0.00
Seed 1	241.2	-0.04
Seed 2	241.7	-0.04
Seed 3	243.1	-0.03
Seed 4	245.7	-0.02

C Verification of the sensitivities

For verification of the implemented sensitivities, every objective chain was compared with central finite differences on a perturbed lattice state. Mechanical, thermomechanical, deteriorated, and cultured compliances were differentiated through the phase, thickness, and mask channels at a solver tolerance of 10^{-10} . The smallest relative errors over the step sweep from 10^{-3} to 10^{-7} are listed in Table 13. The thermally accelerated deterioration chain was verified separately using a nonuniform temperature field, for which best-step relative errors of $1.6 \cdot 10^{-8}$, $5.3 \cdot 10^{-9}$, and $1.8 \cdot 10^{-6}$ were obtained for the thickness, mask, and phase channels, respectively. By tightening the linear-solver tolerance from 10^{-4} to 10^{-10} , the corresponding verification error was reduced from $1.4 \cdot 10^{-3}$ to $1.6 \cdot 10^{-8}$. The remaining discrepancy was therefore attributable to the forward and adjoint residuals.

Table 13: Overview of the sensitivity verification by central finite differences for each objective chain and control channel at a solver tolerance of 10^{-10} .

Chain	t	m	$\delta\varphi$
Mechanical	$2.2 \cdot 10^{-9}$	$1.3 \cdot 10^{-9}$	$2.1 \cdot 10^{-7}$
Thermomechanical	$6.0 \cdot 10^{-9}$	$1.0 \cdot 10^{-9}$	$1.3 \cdot 10^{-7}$
Deteriorated	$8.9 \cdot 10^{-10}$	$2.6 \cdot 10^{-10}$	$9.0 \cdot 10^{-9}$
Cultured	$9.2 \cdot 10^{-9}$	$3.7 \cdot 10^{-9}$	$1.8 \cdot 10^{-8}$

D Growth model equations and parameters

The growth model represents an idealized approximation of chondrocyte ingrowth under the geometric conditions of the optimized structure. The underlying parameters can, in general, be adapted to the applied cell line, culture medium, and environmental conditions. In the present implementation, the nutrient concentration is described by the stationary pore-space balance

$$\nabla \cdot (\mathcal{D} \nabla c) = \frac{rc}{K_c + c}. \quad (8)$$

Unit concentration is imposed on the outer boundary for every culture condition. The diffusivity remains high within the pore space and low within the solid fraction. Static and perfused cultures differ in the uptake parameter, while the Darcy wall-shear stimulus is additionally introduced under perfusion. The geometric clearance stimulus is derived according to

$$s_{\text{gap}} = \exp[-\langle \delta - t \rangle_+ / \ell_{\text{gap}}],$$

while the strain stimulus follows as

$$s_{\text{mech}} = \frac{e}{e_{\text{opt}}} \exp\left(1 - \frac{e}{e_{\text{opt}}}\right), \quad (9)$$

in which e is normalized by its initial 98th percentile. The regulatory input is restricted according to the nutrient concentration and the pore space obtained from the optimized geometry, while the mechanical channels are added to an unstimulated baseline. The matrix fraction evolves as

$$\frac{\partial \varphi}{\partial t_c} = \Lambda^2 k_{\text{syn}} n s \left(1 - \frac{\varphi}{\varphi_{\text{max}}} \right), \quad (10)$$

with

$$s = s_{\text{acc}} s_{\text{gap}} \frac{w_0 + \sum_i w_i s_i}{w_0 + \sum_i w_i}. \quad (11)$$

The scale of the synthesis rate and the duration of culture are combined in the parameter Λ^2 .

The composite modulus is

$$E = E_{\text{min}} + [\rho_s + \varphi^{q_{\text{neo}}} (1 - \rho) \eta] \Delta E, \quad (12)$$

with $\Delta E = E_0 - E_{\text{min}}$. Scaffold resorption is represented by

$$D_{\text{res}} = D_{\text{max}} [1 - \exp(-(\kappa_{\text{res}} c)^2)], \quad (13)$$

and the resulting reduction is applied to the scaffold contribution while the neotissue accumulates. The initial cell density is $n_0 = 0.15 n_c$ with carrying capacity $n_c = 1 - \rho$, and the explicit cell update is given by

$$\partial_{t_c} n = \nabla \cdot (\mathcal{D}_n n_c \nabla n) + p_n c n (1 - n/n_c) - d_n (1 - c) n, \quad (14)$$

with zero flux on the outer boundary. The diffusivity and uptake fields are $\mathcal{D} = \mathcal{D}_s + (1 - \rho)(1 - b_\varphi \varphi)(1 - \mathcal{D}_s)$ and $r = r_0(1 - \rho)(1 + a_\varphi \varphi)$, respectively. The saturating nutrient term is approximated by two fixed-point iterations. The Darcy permeability is $k = 10^{-6} + (1 - \rho)^3$ with a unit pressure difference between the left and right boundaries. The wall-shear surrogate $\tau = k |\nabla p_D| / \sqrt{k}$ is normalized by its 98th percentile. The accessibility stimulus is $s_{\text{acc}} = c$. The swelling force is assembled as $f_{\text{sw}} = \sum_e E_{t,e} \varepsilon_{\text{sw}} \varphi_e \int \mathbf{B}^T \hat{\mathbf{D}} [1, 1, 0]^T dV$, where $E_{t,e}$ denotes the tissue share of the composite modulus. All state fields are advanced over three increments with eight cell substeps per increment.

Table 14: Overview of the nondimensional growth model parameters, with lengths given in element edges.

Symbol	Meaning	Value
φ_{max}	Tissue fraction saturation	0.9
q_{neo}	Matrix-stiffness exponent	2
η	Neotissue-to-scaffold modulus	0.1
ℓ_{gap}	Clearance decay length	2
e_{opt}	Optimal amplitude	0.6
w_0, w_i	Baseline, stimulus weights	0.5
$\mathcal{D}_n$	Cell mobility	0.35
p_n, d_n	Proliferation, death rate	0.9, 0.5
$n_0/(1-\rho)$	Seeded cell density	0.15
k_{syn}	Synthesis rate	1.0
r_0	Uptake, static, perfused	1/16, 1/400
κ_{res}	Resorption severity	0.8
Δt_c	State increment, cell substep	1/3, 1/24
ε_{sw}	Swelling strain	0.06
K_c	Uptake half-saturation constant	0.3
$\mathcal{D}_s$	Diffusivity in the solid	10^{-3}
b_φ, a_φ	Transport blocking, uptake demand	0.7, 2.0
D_{max}	Resorption saturation	0.8

E Epoch-dependent sensitivity fields

The density and the five sensitivity-related fields at every tenth epoch of the reference optimization are shown in Fig. 18. For the diverging derivative fields, the same symmetric scale as in Fig. 2 is used, thereby allowing the localization and relative persistence of the channels to be compared over the complete continuation.

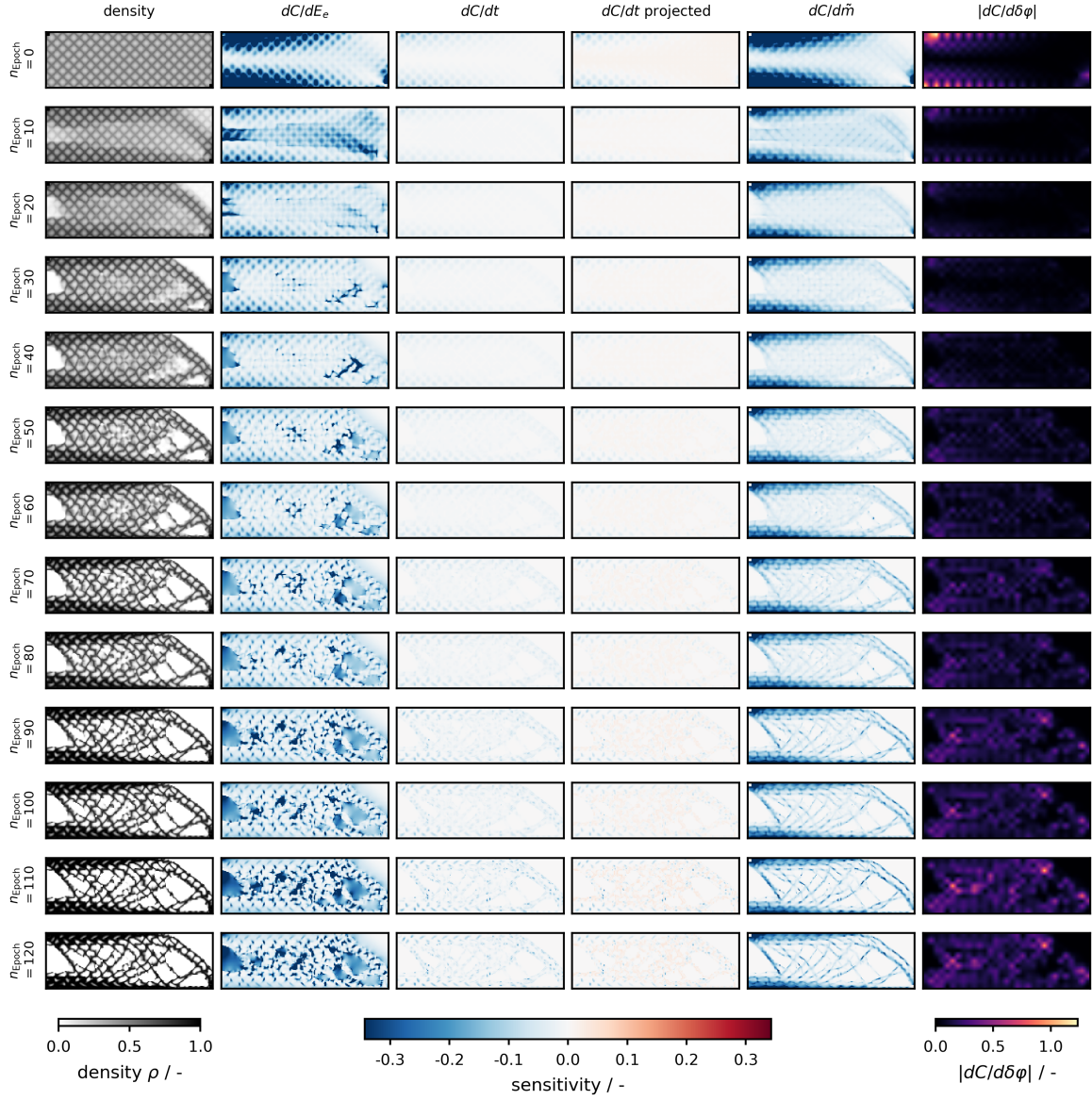


Figure 18: Overview of the evolution of the sensitivity fields over the optimization run.

F Notation

The symbols used below are summarized for reference. On the design side, $\delta\varphi$ denotes the two phase corrections, a the control mapped to the half thickness t , and m the control mapped to the projected mask $\tilde{m}$, all controls being collected in the control vector $\mathbf{p}$. The synthesized geometric fields are collected in $\mathcal{G}$. The lattice function is f , its first-order distance surrogate is δ , and the phase-map Jacobian is $\mathbf{J}$. The Basquin slope is denoted by m_B to distinguish it from the macro-topology control m . On the physical side, μ denotes a generic material-property field, E the element modulus, u the displacement, K the stiffness matrix, b the excitation, and C the compliance. The matrix $\mathbf{B}$ is the strain-displacement matrix and $\hat{\mathbf{D}}$ the normalized plane-stress constitutive matrix. The fields u_{th} and u_b denote the pure-thermal and pure-mechanical displacement solutions, respectively, and T_e is the absolute temperature of element e . The aged and cultured compliances are denoted by C_a and C_g , respectively. The interpolation exponents are $p_{msk} = 3$ and $q_{neo} = 2$, while $q_{lat} = 1$ in the principal campaigns and $q_{lat} = 3$ in the penalized-interface benchmark. The state variables are the damage D , cell density n , and neotissue fraction φ , driven by exposure fields s and represented generically by z . Transport is described by the concentration c , diffusivity $\mathcal{D}$, uptake r , and half-saturation constant K_c . The culture time is t_c , the Darcy pressure is p_D , and the cell mobility is $\mathcal{D}_n$. All lengths are given in element edges and all moduli relative to $E_0 = 1$ so that the resulting quantities are therefore reported as nondimensional values.

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